

Optional Lab

Custom-Agent Feature Delivery

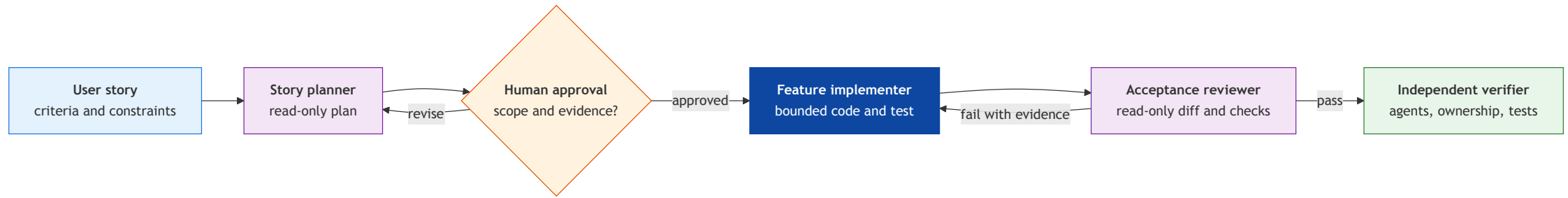
30 minutes | Individual or pairs

Mission

Create and run three project-scoped Claude Code agents to deliver a notification-preference user story.

- **Tools:** Installed Claude Code, Git, Python 3, and `unittest`
- **Workflow:** Plan, approve, implement, review, and verify
- **Output:** Agent definitions, bounded production diff, and passing evidence
- **Source:** `day1/extensions/custom-agent-feature-delivery/`

Evidence-Gated Agent Chain



Create the Workspace

From the optional lab directory:

```
python3 start/setup_lab.py agent-lab-workspace  
cd agent-lab-workspace  
python3 -m unittest tests.test_renderers  
python3 -m unittest tests.test_preferences  
claude
```

Expect the renderer baseline to pass and four preference tests to fail.

30-Minute Run of Show

Time	Checkpoint
0-4	Create the workspace and read the failures.
4-10	Create and inspect three project agents.
10-14	Invoke the planner and approve its boundary.
14-22	Invoke the implementer and inspect its diff.
22-27	Invoke the reviewer and inspect its evidence.
27-30	Run the independent gate and share a decision.

Create Three Project Agents

Use the complete creation prompt in the README. Store definitions under

```
.claude/agents/
```

Agent	Model	Tools	Responsibility
story-planner	Haiku	Read, Glob, Grep	Bound the story and checks
feature-implementer	Sonnet	Read, Glob, Grep, Edit, Bash	Change owned code and test it
acceptance-reviewer	Sonnet	Read, Glob, Grep, Bash	Review without editing

Reject a planner or reviewer with editing access.

Approve the Plan Before Coding

Ask the main session to delegate `story.md` analysis to `story-planner` and stop after returning the plan.

Approve only when the plan:

1. assigns only `preferences.py` and `router.py` ;
2. treats tests as immutable;
3. covers every preference case;
4. names focused and full verification commands.

Revise weak assumptions before invoking the implementer.

Implement, Then Review Independently

Invoke `feature-implementer` on the approved plan. Inspect the real diff and focused result.

Then invoke `acceptance-reviewer` to:

- inspect `git diff` from `lab-base`;
- run focused and full checks;
- return PASS or FAIL for every criterion;
- provide exact failure evidence without editing.

Route failures back to the implementer for the smallest correction.

Verify and Share Evidence

```
python3 verify_lab.py  
git diff --name-only lab-base  
git ls-files --others --exclude-standard
```

Expected marker: **CUSTOM AGENT LAB VALID**

Show the three definitions, five allowed changed files, and verifier output. Explain one assumption you rejected, revised, or verified independently.

Definition of Done and Help Order

- Three valid project agents use role-appropriate models and tools.
- The transcript shows three separate delegations.
- Human approval occurs before implementation.
- Only agent definitions, `preferences.py`, and `router.py` change.
- Focused, full, and independent checks pass.

If blocked: failing test, agent definition, delegation evidence, diff boundary, exact repair prompt, then `solution/`.

Return for the Debrief

Bring the agent chain, the bounded diff, and one evidence-based decision.